GGG-W-646
Interim Amendment-4
August 31, 1962

FEDERAL SPECIFICATION

WRENCH, OPEN END, RATCHET (TAC PATTERN), FOR TUBE FITTINGS, ELECTRIC CABLE TERMINALS AND STUFFING TUBE GLAND NUTS

This Interim Amendment was developed by the Standardizatio.. Division, Federal Supply Service, General Services Administration, Washington 25, D. C., based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized Federal agencies to use this Interim Amendment as a valid exception to Federal Specification GGG-W-646 dated July 13, 1954 and Amendment-3 dated January 15, 1962.

Page 1, paragraph 1.2, line 9: After "Open end" add "(electronic cable fitting wrench)".

Page 1, paragraph 2.1: Add the following:

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

Military Specification:

MIL-H-15424 - Hand Tools; Packaging of

Page 2, paragraph 2.1: Under Military Standards add:

MIL-STD-129 - Marking for Shipment and Storage.

Page 2, paragraph 2.1: Delete parenthetical paragraphs and substitute the following:

FSC 5120

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Page 2, paragraph 2.2: Delete in its entirety.

Page 2, paragraph 3.2.1, line 3, after "cracks," add "blowholes,".

Page 2, paragraph 3.2.2, sentences 2 and 3: Delete and substitute: "Unless otherwise specified (see 6.1), wrenches shall have one or a combination of the coatings listed herein; if chromium and nickel-plate coatings are specified but are restricted by Government order, one of the other coatings listed will be acceptable. The rotating member or ratchet of the wrench head and the hexagon drive of the internal wrenching attachments shall be phosphate or oxide coated."

Page 3, paragraph 3.5.1, third line: Between "member" and "and" add "a pawl release mechanism."

Page 5, table II, column 5, item 10: Delete ".995" and substitute ".955".

Page 6, table II, column 5, item 11: Delete "2.470" and substitute "2.473".

Page 7, table III: Delete and substitute new table III.

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TABLE III.—*Type I, class 1, style B, form B, open ratchet, crowfoot single-purpose; open-end (electronic cable fitting wrench) wrench heads*

Component number	Rated torque	Nominal wrench size	GG[1]	Dimensions						
				NN		JJ[1]	LL +0.015 −.005	MM +0.005 −.000	KK	
				Minimum	Maximum				Degrees	Minutes
	Inch-pounds	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*		
33	150	3/8	1.380	0.378	0.386	1.820	0.322	0.254	7	30
34	180	7/16	1.380	.440	.448	1.820	.322	.254	7	30
35	200	1/2	1.380	.504	.512	1.820	.322	.254	7	30
36	225	9/16	1.630	.566	.575	2.200	.352	.381	7	30
37	380	5/8	1.933	.629	.638	2.433	.362	.381	7	30
38	420	11/16	2.053	.692	.701	2.593	.372	.381	7	30
39	460	3/4	2.173	.755	.765	2.733	.400	.381	7	30
40	520	13/16	2.301	.818	.828	2.901	.410	.381	7	30
41	675	7/8	2.421	.880	.890	3.021	.420	.381	7	30
42	730	15/16	2.561	.944	.955	3.161	.425	.381	7	30
43	840	1	2.685	1.006	1.017	3.285	.435	.381	7	30
194	930	1 1/16	2.807	1.068	1.080	3.407	.460	.381	7	30
195	930	1 1/8	2.807	1.132	1.145	3.407	.460	.381	7	30
196	970	1 3/16	2.987	1.194	1.207	3.687	.490	.508	7	30
44	970	1 1/4	2.987	1.257	1.270	3.687	.490	.508	7	30
197	1030	1 5/16	3.107	1.320	1.334	3.807	.490	.508	7	30
45	1030	1 3/8	3.107	1.383	1.397	3.807	.490	.508	7	30
198	1100	1 7/16	3.307	1.446	1.460	4.007	.490	.508	7	30
46	1100	1 1/2	3.307	1.508	1.523	4.007	.490	.508	7	30
199	1175	1 9/16	3.467	1.571	1.586	4.167	.490	.508	7	30
47	1175	1 5/8	3.467	1.634	1.649	4.167	.490	.508	7	30
200	1250	1 11/16	3.627	1.696	1.711	4.327	.490	.508	6	0
48	1250	1 3/4	3.627	1.759	1.775	4.327	.490	.508	6	0
201	1300	1 13/16	3.692	1.822	1.838	4.392	.490	.508	6	0
49	1300	1 7/8	3.692	1.885	1.901	4.392	.490	.508	6	0
202	1350	1 15/16	3.807	1.948	1.964	4.527	.490	.508	6	0
50	1350	2	3.807	2.011	2.028	4.527	.490	.508	6	0
203	1400	2 1/16	3.957	2.074	2.092	4.707	.570	.508	6	0
51	1400	2 1/8	3.957	2.136	2.155	4.707	.570	.508	6	0
204	1500	2 3/16	4.117	2.200	2.219	4.867	.580	.508	5	0
52	1500	2 1/4	4.117	2.262	2.281	4.867	.580	.508	5	0
205	1500	2 5/16	4.440	2.324	2.344	5.190	.580	.508	5	0
53	1500	2 3/8	4.440	2.388	2.403	5.190	.580	.508	5	0
206	1500	2 7/16	4.607	2.453	2.491	5.367	.600	.508	5	0
54	1500	2 1/2	4.607	2.513	2.534	5.367	.600	.508	5	0
207	1500	2 9/16	4.780	2.576	2.587	5.540	.680	.508	5	0
55	1500	2 5/8	4.780	2.639	2.660	5.540	.680	.508	5	0
208	1500	2 11/16	4 957	2.701	2.722	5.737	.750	.508	5	0
56	1500	2 3/4	4.957	2.766	2.787	5.737	.750	.508	5	0
209	1500	2 13/16	4.957	2.827	2.849	5.737	.750	.508	5	0
57	1500	2 7/8	4.957	2.890	2 912	5.737	.750	.508	5	0
210	1500	2 15/16	4.957	2.954	2.977	5.737	.750	.508	5	0
58	1500	3	4.957	3.016	3.039	5.737	.750	.508	5	0

[1]Nominal dimension, held within Drop-forging Association tolerances.

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Page 8, paragraph 3.5.1.2.2, line 1: After "open end" add "(electronic cable fitting wrench)".

Page 8, figure 3, title: After "open end" add "(electronic cable fitting wrench)".

Page 8, paragraph 3.6.1: Delete the last two sentences and substitute: "The sockets shall be made of steel and shall be free of all pits, nodules, burrs, cracks, blowholes, and other detrimental defects. The nut engaging surfaces shall be smooth. The sockets shall have a Rockwell hardness of 40 to 52 on the "C" scale."

Page 10, table V, column 7, item 20: Delete "1.825" and substitute "1.875".

Page 12, table VI, columns 5,6,8, and 10: Delete and substitute:

AA		CC ± .032	EE ± .032
Minimum	Maximum		
Inches	Inches	Inches	Inches
1.056	1.066	1.437	1.109
1.150	1.160	1.562	1.109
1.275	1.285	1.687	1.109
1.400	1.411	1.812	1.250
1.468	1.474	1.875	1.250
1.681	1.693	2.187	1.265
1.806	1.818	2.312	1.265
1.900	1.910	2.437	1.281
1.962	1.975	2.500	1.281
2.213	2.228	2.812	1.281
2.275	2.290	2.875	1.359
2.369	2.385	3.000	1.359
2.494	2.510	3.125	1.359
2.838	2.856	3.500	1.375
2.994	3.013	3.625	1.390
3.119	3.139	3.750	1.453
3.306	3.326	4.000	1.453
3.463	3.484	4.187	1.468
3.587	3.609	4.312	1.468
3.775	3.798	4.500	1.468

Page 12, table VI, footnote: Delete.

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Page 13, table VIII, column 3, heading: Delete "9" and substitute"8".

Page 16, under "Nomenclature" column, line 2: Delete "standard" and substitute "thin".

Pages 19 and 20, paragraphs 4.1 thru 4.3, and tables XI and XII - Delete and substitute:

4.1 Inspection responsibility. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. - Unless otherwise specified therein, sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in Standard MIL-STD-105.

4.2.1 Inspection of materials and components. - In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2.2 Inspection of the end item (wrench). - Inspection of the wrenches shall be performed as specified in the following paragraphs. A lot shall consist of wrenches of the same type, class, style and form. The sample unit shall be one completely fabricated and assembled wrench.

4.2.2.1 Visual examination. - Each sample unit shall be visually examined for the defects set forth below. The inspection level shall be level II, with as acceptable quality level (AQL) of 4.0 defects per hundred units.

Examine	Defect
Finish	Protective coating or plating, as applicable, missing or not as required.
	Not smooth, uniform or adherent, peeling, tacky.
	Not free of burrs, nodules, pits, nicks, scale or other imperfections.

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Examine	Defect (Cont.)
Material	Not of uniform quality, cracks, excessive porosity, not suitable for intended purpose.
Design	Not type, class, style, form or size specified.
Construction and Workmanship	Parts missing, damaged or not properly assembled.
	Ratchet mechanism inoperative, enclosure plate not provided for accessibility in repairs (type 1 wrench head only).
	Spring-loaded ball missing, or will not function as intended (type 1, class 1, style A only).
	Any operation missing or incomplete.
Marking	Illegible, incomplete, not permanent, missing.

4.2.2.2 Dimensional examination. - Each sample unit shall be examined for dimensions specified in applicable paragraphs and tables. Any deviation therefrom shall constitute a defect. The inspection level shall be level L 6 with an AQL of 4.0 defects per hundred units.

4.2.2.3 Testing. - Each sample unit shall be tested in accordance with 4.4.1 and 4.4.2. For the torque test, the inspection level shall be L 6 with an AQL of 4.0 defects per hundred units. For the hardness test, lots identifiable as having been heat-treated as a single batch or under the same process and temperature control, the inspection level shall be L 4 with an AQL of 1.5 defects per hundred units. For lots not identifiable as such the inspection level shall be L 7 with an AQL of 1.5 defects per hundred units. Testing for hardness may be performed at an appropriate intermediate station prior to finish.

4.2.2.4 Examination of end item (sets). - Visual examination of the sets shall be made in accordance with the list of defects set forth below, which shall apply to all components of the set, including the container. A lot shall be composed of wrenches, attachments and adapters that have satisfactorily passed the visual and dimensional examinations and applicable tests under 4.4. The sample unit shall be one completely fabricated and assembled set. The inspection level shall be L 6 with an AQL of 6.5 defects per hundred units.

Examine	Defect
Finish (container & tools)	Scratches, chips or mars affecting appearance.
	Fins, burrs, or sharp edges.
	Box not completely coated inside and outside as required.
	Box not provided with handle (when required).

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Examine	Defect (Cont.)
Construction and Workmanship (container & tools)	Parts missing, or not type, class, style, form or size specified.
	Wrench sets not compactly housed in container.
	Components not held securely in place.
	Components not readily attachable to or detachable from their handles or attachments.
	Component bent, cracked, sprung or otherwise damaged.
Marking	Illegible, incomplete, not permanent or missing.

4.3 Inspection of preparation for delivery. - Inspection to determine compliance with preparation of delivery requirements shall be performed in accordance with Specification MIL-H-15424.

Page 20, paragraph 4.4.1 - Delete the phrase "in accordance with table XII" of the first sentence.

Delete last two sentences.

Page 20, paragraph 4.4.2 - Delete and substitute:

4.4.2 Hardness. - Testing to determine compliance with requirements specified in 3.6.1 shall be performed in accordance with Method 243.1 of Federal Test Methods Standard No. 151.

Pages 20 and 21: Delete section 5 and substitute new section 5:

5. PREPARATION FOR DELIVERY

(Civil agencies. - Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.

5.1.1 Level A. - Level A preservation and packaging shall be in accordance with level A of Military Specification MIL-H-15424.

5.1.2 Level C. - Level C preservation and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 Level A. - Level A packing shall be in accordance with level A packing of Military Specification MIL-H-15424.

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5.2.2 Level B. - Level B packing shall be in accordance with level B packing of Military Specification MIL-H-15424.

5.2.3 Level C. - Level C wrenches preserved and packaged in accordance with level A or C as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. This level in general shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements.

5.3 Marking.

5.3.1 Civil agencies. - Unless otherwise specified (see 6.1), marking for shipment shall be in accordance with Federal Standard No. 123.

5.3.2 Military agencies. - In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Military Standard MIL-STD-129.

Page 21, paragraph 6.1, item (f): Delete and substitute:

(f) Whether preservation and packaging should be level A or C and whether packing should be level A, B, or C (see section 5).

Page 21, paragraph 6.1: Add as item (g):

(g) Marking, if different (see 5.3.2). Pages 21 and 22, section 7: Delete:

Military Custodians:

Army - Q
Navy - ShMC
Air Force - MOA

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